Math 205A - Fall 2014
Homework #2
Solutions

Problem 1: Show that there exist two countable sub-collections $\mathcal{F}_1$, $\mathcal{F}_2$ of pairwise disjoint intervals, such that $\mathcal{F}_1 \cup \mathcal{F}_2$ covers A.

We'll first cover $A \cap (0,1)$, then extend to the whole $\mathbb{R}$, so assume for now that $A \subset (0,1)$. The strategy is to initially cover A inductively by a countable collection of intervals that are *not necessarily disjoint*; afterwards we'll rearrange these intervals into 2 sub-collections, each of them disjoint.

Step 1. Constructing a countable cover for A.

Let $A_1 = A$, $\mathcal{G}_1 = \{I \in \mathcal{F} : I \subset (0,1)$ and center of I is in $A_1\}$, $\alpha_1 = \sup\{|I| : I \in \mathcal{G}_1\} \leq 1$. If $\alpha_1 = 0$, there's nothing to prove. Otherwise, choose $I_1 \in \mathcal{G}_1$ centered at $x_1 \in A_1$ with $|I_1| > 3/4\alpha_1$.

Given $A_i, \mathcal{G}_i, I_i$ for $i = 1, \dots, n-1$, define $A_n = A \backslash \cup_{i=1}^{n-1} I_i$, $\mathcal{G}_n = \{I \in \mathcal{F} : I \subset (0,1)$ and center of I is in $A_n\}$, and $\alpha_n = sup\{|I| : I \in \mathcal{G}_n\}$. If $\alpha_n = 0$, then $A \subseteq \cup_{i=1}^{n-1} I_i$ (remember, the intervals are non-degenerate). Otherwise again pick $I_n \in \mathcal{G}_n$ centered at $x_n \in A_n$ with $|I_n| > 3/4\alpha_n$.

First, $\alpha_n \to 0$: In deed, if $\alpha_n = 0$ for some n, we're done. Otherwise $\alpha_{n+1} \leq \alpha_n$, so say $\alpha_n \downarrow \alpha \geq 0$. If $m > n$, then $x_m \notin I_n$, so $|x_m - x_n| \geq |I_n|/2 \geq 3/8\alpha_n \geq 3/8\alpha$. Therefore we have an infinite sequence x_n of elements in (0,1) with distance between any two $\geq 3/8\alpha$, which can only happen if $\alpha = 0$.

Now we claim that $A \subseteq \cup I_n$. If not, let $x \in A \backslash (\cup I_n)$, and $I \subset (0,1)$ any interval in $\mathcal{F}$ centered at x. Since $x \in A \backslash (\cup I_n)$, then for all n, $x \in A_n$, so $I \in \mathcal{G}_n$, therefore $|I| \leq \alpha_n$. But $\alpha_n \to 0$, hence $|I| = 0$, contradicting the non-degeneracy assumption.

Step 2. Getting rid of the 'redundant intervals'.

We'll now get a new sub-collection I_n' that has 'less' overlaps than the original one as follows: If $A \subset \cup_2^\infty I_n$, let $I_1' = \emptyset$, otherwise $I_1' = I_1$. In step n, if $A \subset (\cup_1^{n-1} I_i') \cup ((\cup_{n+1}^\infty I_i))$, let $I_n' = \emptyset$, otherwise $I_n' = I_n$. Then $A \subset \cup I_n'$, because by construction every point in A is contained only in finitely many of the $I_n's$ (if $x \in I_k$ for the first time, then $dist(x, x_l) > 0$ for $l > k$, and $|I_n| \downarrow 0$), so we could not have removed all of them.

What we achieved this way is that at most two of the non-empty I_n''s overlap at any point, because if I_i, I_j, I_k all intersect, then one of them is included in the others, say I_i. But then $I_i' = \emptyset$, contradicting the non-emptiness.

Step 3. Obtaining $\mathcal{F}_1$ and $\mathcal{F}_2$.

There are many ways to do this, but one nice way is using graph theory: let each I_n' be a vertex of a (possibly infinite) graph, and connect two vertexes iff the corresponding intervals overlap. By the remark above, this can have no cycles, so it's a tree, and hence bipartite. This means that the vertices can be arranged into two sets $S_{1,2}$, each of them with no edges in between. Then put the intervals belonging to the set S_i into $\mathcal{F}_i$!

Step 4. Covering A (not only $A \cap (0,1)$).

For each $n \in \mathbb{Z}$, pick an interval J_n of radius $< 1/2$ if $n \in A$, otherwise do nothing. $\mathbb{R} \backslash (\cup J_n)$ is a disjoint union of open intervals, each $\subseteq (n, n+1)$ for some n, so pick the disjoint collections $\mathcal{F}_1^n, \mathcal{F}_2^n$, also disjoint from the J_n's. Then $\mathcal{F}_1 = \cup_n \mathcal{F}_1^n \cup \{J_n\}_n$ and $\mathcal{F}_2 = \cup_n \mathcal{F}_2^n /$ □

Problem 2: Given an open set U, prove that there exists a countable disjoint collection of intervals $\mathcal{J}$ such that $\cup_{I \in \mathcal{J}} I \subset U$ and $\mu(A \backslash (\cup_{I \in \mathcal{J}} I)) = 0$.

Using problem 1, find collections $\mathcal{F}_1$, $\mathcal{F}_2$ (each disjoint) of intervals in $U_1 = U$ that cover $A \cap U$. Then $\mu(A \cap (\cup_{\mathcal{F}_i} I) \geq \mu(A \cap U)/2$, say for $\mathcal{F}_1$. Then there exist finitely many disjoint intervals $I_1, \dots, I_{n_1}$ with $\mu(A \cap (\cup_1^{n_1} I_i)) \geq \mu(A \cap U)/3$. Let $U_2 = U_1 \backslash (\cup_1^{n_1} I_i)$. This is again open, and $\mu(A \cap U_2) \leq 2/3\mu(A \cap U)$. Repeat the same with U_2 instead of U_1 to get $I_{n_1+1}, \dots, I_{n_2}$ all disjoint from the previous I_i's and each other, and with $\mu(A \cap (U_2 \backslash (\cup_{n_1+1}^{n_2} I_i))) \leq 2/3\mu(A \cap U_2) \leq (2/3)^2 \mu(A \cap U)$. Continuing this way, we obtain a disjoint sequence $I_1, I_2, \dots$ that satisfies $\mu(A \backslash (\cup I_i)) \leq (2/3)^n \mu(A \cap U)$ for all n, therefore giving $\mu(A \backslash (\cup I_i)) = 0$. □

Problem 3: Construct a monotone function that is discontinuous on a dense set on $[0,1]$.

Let $q_1, q_2, ...$ be an enumeration of rationals, and let $f(x) := \sum 2^{-n}\chi_{(q_n,\infty)}$. f is monotone, as a sum of increasing functions. Furthermore, it's discontinuous in $\mathbb{Q}$: given n, for all $x > q_n$, $f(x) \geq 2^n + f(q_n)$. □

Problem 4: Show that almost every x lies in at most finitely many of the E_k's.

$$\mu(\{x \text{ lies in infinitely many } E_k\}) = \mu(E^c = \cap_{n=1}^{\infty} \cup_{k\geq n} E_k) = \mu(E^c) = \lim_{n\to\infty} \mu(\cup_{k\geq n} E_k) \leq \lim_{n\to\infty} \sum_{k=n}^{\infty} \mu(E_k) = 0,$$

since $\sum \mu(E_k) < \infty$. Hence, almost every x lies in at most finitely many E_k's. □

Problem 5: **(i)** Show that $\phi_t(g) \to g(0)$.
(ii) How much can we weaken the regularity assumptions on ϕ and g?

(i) Let $M = sup|g(x)| < \infty$ (since g is C^∞ and with compact support), and let $\epsilon > 0$. Then there exists a $\delta > 0$ s.t. $|x| < \delta$ implies $|g(x) - g(0)| < \epsilon$. Now noting that by a change of variables $\int \phi_t(x)dx = 1$, we get:

$$\begin{aligned}
|\phi_t(g) - g(0)| = \; & |\int \phi_t(x)(g(x) - g(0))dx| \leq \int_{|x|<\delta} \phi_t(x)|g(x) - g(0)|dx + \int_{|x|\geq\delta} \phi_t(x)|g(x) - g(0)|dx \\
\leq \; & \epsilon \int_{|x|<\delta} \phi_t(x)dx + 2M \int_{|x|\geq\delta} \phi_t(x)dx \leq \epsilon \int_{\mathbb{R}^n} \phi_t(x)dx + 2M \underbrace{\int_{|y|\geq\delta/t} \phi(y)dy}_{\text{change of variables } x=ty} \\
\leq \; & \epsilon + 2M \int_{|y|\geq\delta/t} \phi(y)dy
\end{aligned}$$

But $\int \phi = 1$ implies that $\int_{|y|\geq\delta/t} \phi \longrightarrow 0$ as $t \to 0$. Hence, $\lim_{t\to 0} |\phi_t(g) - g(0)| \leq \epsilon$. Since this is true for every ϵ, the result follows.

(ii) In the proof above, all we needed was for g to be bounded and continuous, and ϕ continuous. (The integrals all make sense if g and ϕ are continuous, and $\int \phi_t = 1$.) □